

```

Final.ino
1  #include <ESPAsyncWebServer.h>
2  #include <AsyncTCP.h>
3  #include <WiFi.h>
4  #include <Wire.h>
5  #include <Adafruit_INA219.h>
6  #include <AsyncWebsocket.h>
7
8  // WiFi Credentials
9  const char* ssid = "realme85g"; // Replace with your WiFi SSID
10 const char* password = "vedant12"; // Replace with your WiFi password
11
12 // Sensor and Server Objects
13 Adafruit_INA219 ina219; // INA219 Sensor Object
14 AsyncWebServer server(80); // Web Server on Port 80
15 AsyncWebsocket ws("/ws"); // WebSocket at endpoint /ws
16
17 // Function to send sensor data to all connected WebSocket clients
18 void sendSensorData() {
19     float busVoltage = ina219.getBusVoltage_V(); // Voltage at load (V)
20     float current_mA = ina219.getCurrent_mA() * 1.25; // Current (mA) with calibration
21     float power_mW = ina219.getPower_mW(); // Power (mW)
22     float shuntVoltage_mV = ina219.getShuntVoltage_mV(); // Shunt Voltage (mV)
23     float loadVoltage = busVoltage + (shuntVoltage_mV / 1000.0); // Load Voltage (V)

```

Fig. 2: Snippet of the main source code for Wi-Fi configuration

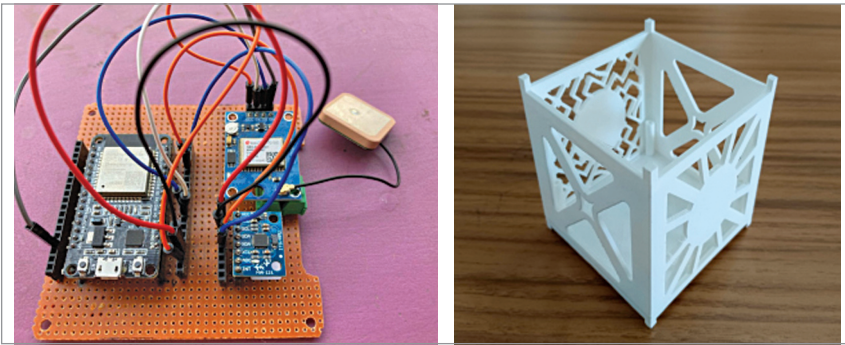


Fig. 3: Assembled circuit and 3D printed body

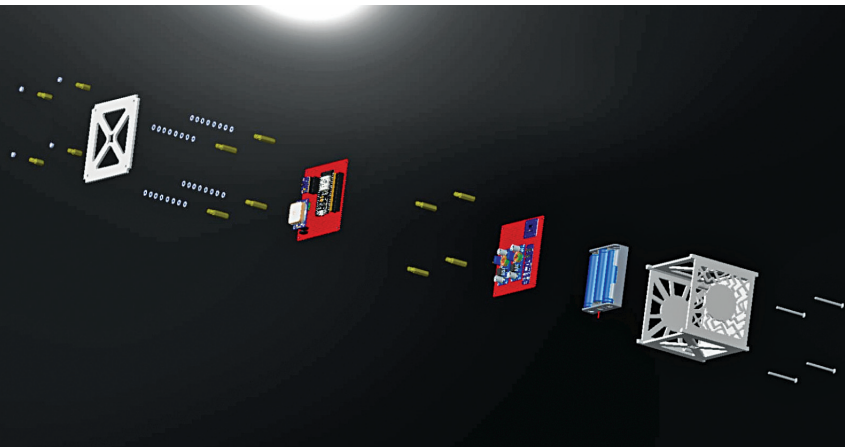


Fig. 4: Satellite payload assembling

modules, which enables viewing their output on the serial monitor. The main transmitter code requires modification to enable remote transmission of the payload data to

the webpage.

A battery management system module supports the power system. Its output connects to a buck converter, which supplies power to

BILL OF MATERIALS

Components	Quantity	Details
Dotted PCB/perf board	1	Baseboard
ESP32 development board	1	Controller
MPU6050 gyro sensor module	1	IMU
NEO-6M GPS module	1	Positioning
INA219 current sensor	1	Monitoring
Buck converter	2	Regulator
Screw terminals	4	Connectors
Pin headers	15	Interface
Pin sockets	30	Mounting
18650 batteries	3	Power
3D printing files for CubeSat frame	1	Structure
BMS board	1	Battery management

Note: This is not an actual satellite, but rather a low-orbit satellite model designed for educational purposes that operates within a Wi-Fi range only.

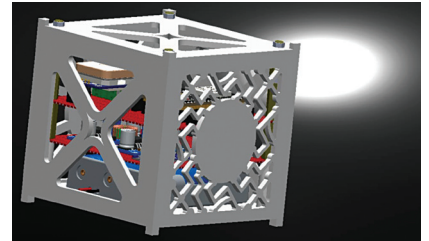


Fig. 5: Assembled satellite box

the OBC. A second buck converter powers additional payloads. The INA219 current sensor and MPU6050 are both connected to the ESP32's I2C bus, while the GPS module connects via the ESP32's USART pins.

Software

Three source code files are included. The main file hosts a webpage and sends real-time power readings to it. The remaining two are test files that display data from the GPS and MPU6050 sensors via the serial monitor.

The main code can be modified to incorporate the payload data into the hosted webpage by referencing the test codes. Additional sensors can also be connected to extend the model further.